MAHARSHI OZA

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PROFESSIONAL SUMMARY

Verification and validation engineer with experience across ADAS perception (camera and LiDAR-based modules at Magna Electronics) and hardware verification (SystemVerilog/UVM at Maxgen Technologies). Designs test plans and requirements-based acceptance checks, builds Python and C++ test automation and reusable test frameworks, and analyzes large test datasets to root-cause failures. MS in Computer Engineering, NC State.

CORE COMPETENCIES

V&V & Test Engineering: Test Planning, Requirements-Based Testing (SRD/SRS), Acceptance Criteria & KPIs, Functional & Regression Testing, Reusable Test Frameworks, Root-Cause Analysis, Test Reporting
Python & Data Analysis: Python test automation, data acquisition and post-processing, report parsing, NumPy, PyTorch; automated multi-run test flows producing 45-metric datasets
Sensors & Perception: LiDAR- and Camera-based Object Detection (FOD, NFCD), Lane Detection, Camera Calibration (Intrinsic/Extrinsic), Fisheye Models, Bird's-Eye View, ONNX Inference
Hardware & Embedded: ECU Bench Bring-up, JTAG & GDB Debug, CAN Bus (Vector CANoe/CANalyzer), Embedded C/C++, MISRA C/C++, SystemVerilog/UVM, SVA, Linux
Tools & Practice: Git, Makefile, Tcl, Shell, OpenCV, QuestaSim, Xcelium, VCS, Code/Branch/Line Coverage

PROFESSIONAL EXPERIENCE

RTL Design Verification Engineer | Maxgen Technologies Pvt Ltd., India **Aug 2023 – Aug 2024**

- Implemented an out-of-order scoreboard for an AXI4 memory subsystem in SystemVerilog/UVM, tracking outstanding transactions by ID and checking read/write data, responses and per-ID ordering.
- Developed functional coverage for AXI4 bursts, IDs, out-of-order completion and handshake stalls (over 90%); verified reset, FIFO full/empty and handshake corner cases and debugged regressions in Questa, Xcelium and VCS.
- Maintained UVM environments for Asynchronous FIFO and FPU blocks and contributed to multi-port packet router RTL (FSM control, FIFOs, synchronizers).

Functions Engineer – Vision & Perception | Magna Electronics, India **Dec 2021 – Aug 2023**

- Implemented requirements-based KPI checkers in C++ for the ADAS Lane Marker Detection system, translating SRD/SRS acceptance thresholds into automated pass/fail tests for lane type, color, lateral distance to the rear axle and curvature against ground truth.
- Implemented bird's-eye-view generation in C++ from four surround-view cameras using planar homography with camera intrinsic and extrinsic calibration parameters.
- Validated the classical image-processing and DNN (ONNX) pipelines on recorded camera and CAN datasets with the same acceptance criteria, comparing detection accuracy; supported data annotation and test-set validation.
- Supported ECU bench bring-up and debug: deployed the C++ binary to target hardware, root-caused input-data failures with GDB breakpoints and a JTAG debugger, and analyzed vehicle CAN signals in Vector CANoe/CANalyzer.
- Worked on LiDAR- and camera-based Front Object Detection (FOD) and Near-Field Collision Detection (NFCD) modules.
- Implemented code/branch/line coverage and supported the 32-bit to 64-bit migration (MISRA C/C++).

PROJECTS

Surround-View Calibration, Bird's-Eye View and Lane Measurement Pipeline | C++11, Python, ONNX

- Reverse-engineered a four-camera fisheye recording format with embedded CAN data; built tools to extract frames and CAN logs.
- Implemented the fisheye camera model from calibration data (polynomial distortion, mount tilt/yaw/roll) with exact forward and inverse projection, and a lookup-table stitcher producing a 25.65 mm/px bird's-eye view.
- Wrote a dependency-free ONNX inference engine whose lane outputs match ONNX Runtime exactly, and reported per-frame lane distances from the rear axle (e.g. 1.45 m / 2.17 m, lane width ≈ 3.6 m) to CSV.

Reusable UVM Test Framework for a Write-Back Cache Controller | SystemVerilog, UVM, SVA, Python

- Architected a reusable test environment (CPU, flush and reactive-memory agents, reference-model scoreboard, SVA checks) and a test plan of 7 directed and constrained-random tests including backpressure, race and error-injection scenarios.
- Built a Python regression flow that runs the suite, merges coverage across runs and reports coverage holes; 0 data mismatches across 1,000+ accesses under injected errors.

Automated Test Sweep and Data Analysis – 85-Run Physical Design Study | Python, Makefile, Tcl

- Automated 85 parallel tool runs with Makefiles and Python, parsing 45 metrics per run into one dataset with custom pass/fail checks; analyzed it to find failure boundaries and the best operating point (~5.7× better figure of merit).

Trailer Coupler Height Estimation with a Monocular Fisheye Camera | Magna Electronics | C++, Python, OpenCV

- Undistorted fisheye frames using camera intrinsics, matched features across frames with OpenCV, estimated camera geometry with the 8-point algorithm and RANSAC, and triangulated 3D points to recover coupler height.

Scan Insertion & ATPG Setup: 8-bit Counter (Cadence DFT Fundamentals labs) | Genus, Modus, Verilog, Icarus

- Inserted mux-D scan into an 8-bit counter in Cadence Genus (DFT DRC, one auto-stitched chain) and generated the scan netlist, SCANDEF and Modus ATPG scripts.
- Verified scan shift/capture on the gate-level netlist with a self-checking Verilog testbench (load → capture → unload).

EDUCATION

M.S. Computer Engineering | North Carolina State University, Raleigh, NC | CGPA: 3.7/4 **Aug 2024 – May 2026**
Relevant Coursework: ASIC Verification, Advanced Verification using UVM, Hardware–Software Codesign, Advanced Microprocessor Architecture

B.Tech. Electronics & Communication Engineering | Dharmsinh Desai University, India **Aug 2018 – May 2022**
Relevant Coursework: Digital Image Processing, Digital Signal Processing, Signals and Systems, Object-Oriented Programming with C++, Microprocessors